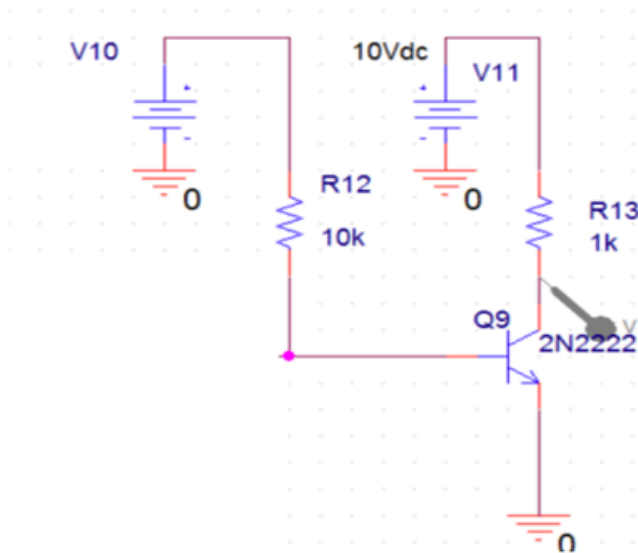


Frank Tamburro

Experiment 1:

Experiment #1: Investigate a Common Emitter Amplifier (Find DC operating point, DC gain)

Construct the following circuit and find the proper DC operating point by adjusting V_{10} and calculate the voltage gain.

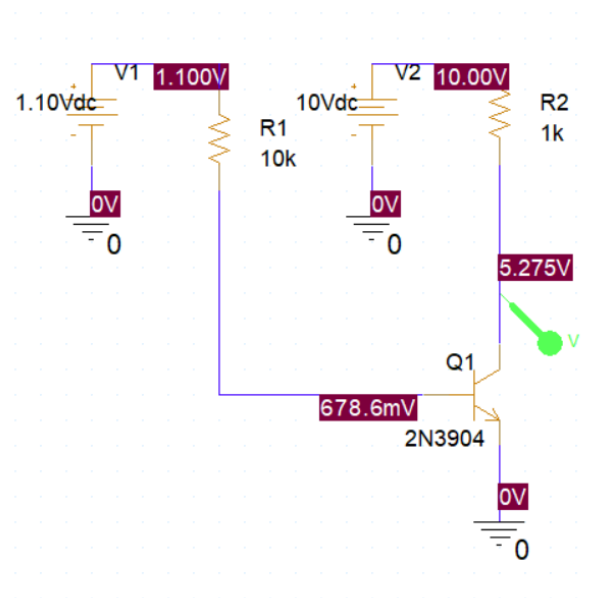


$R_{12}=9.8935\text{KOHM}$

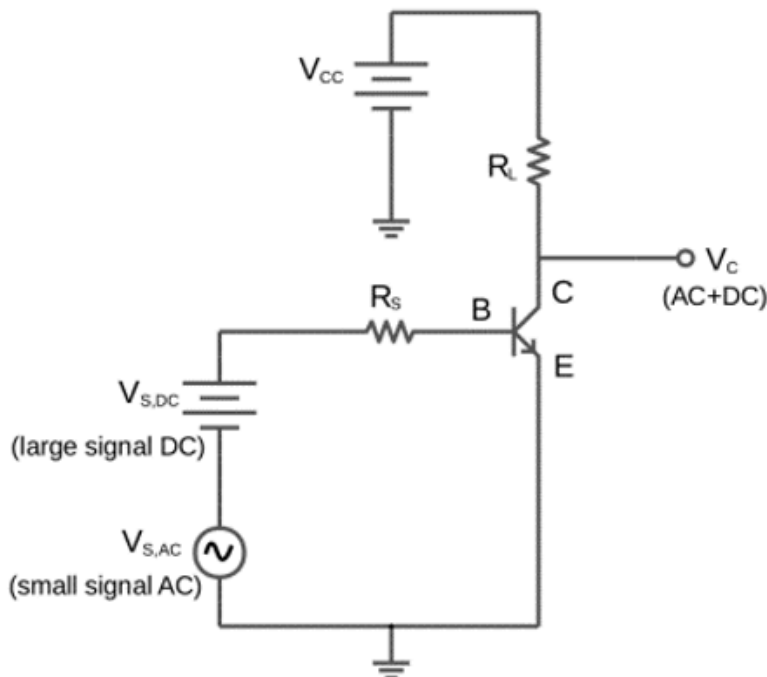
$R_{13}=0.9898\text{KOHM}$

2N3904

$V_{BB}=1.05\text{ Volts}$

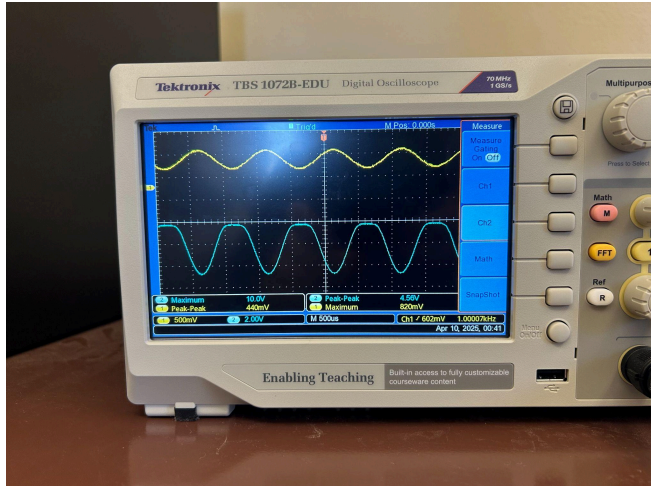


Experiment #2: Investigate a Common Emitter Amplifier (Add a small AC source with a large DC).



1. Construct the circuit in the above figure. Set $R_S=10k\Omega$ and $R_L=1k\Omega$, and power supply $V_{CC}=10V$. As a source at the base utilize a sine wave voltage generator having a DC offset.
2. Set the generator to generate a 1kHz sine wave having as amplitude the lowest value that would produce a sinusoidal signal with no noise (e.g., 200 mVpk-pk) plus a DC offset (indicated in the circuit schematically as $V_{S,AC}+V_{S,DC}$).
3. Monitoring the collector voltage V_C with your oscilloscope, adjust the DC offset so that the average voltage at the output (i.e., collector) is 5 V, this is the DC component of V_C .
4. With the sine wave plus DC offset values, monitor the waveform at the output (i.e. collector) and determine using the oscilloscope the amplitude of the output sine wave (i.e., the AC component of the V_C). If output is not sinusoidal that means the gain or input amplitude are too high and output is distorted. Try a higher R_S resistor and/or lower DC offset until you get a sinusoidal output. If none of that works add a resistor in the emitter as shown in experiment 2.
5. Record the input and output waveforms. Record the amplitudes of both the input and output waveforms. And calculate gain. Adjust the setting of DC offset or R_S to maximize gain. Record all your data, i.e. amplitudes (input & output), DC offset, R_S resistance value.

$R_L=0.9931k\Omega$
 $R_S=9.9201k\Omega$
 $V_{CC}=10V$



Experiment 2:

DC Analysis

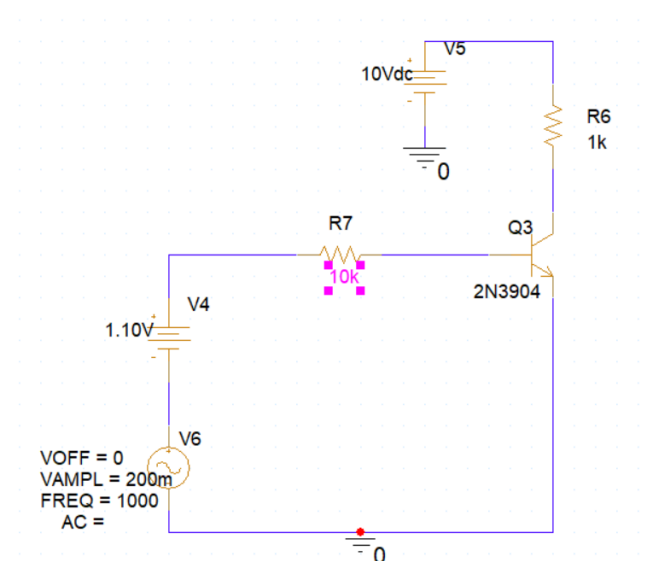
$V_{cc} = 10V$
 $V_c = 5V$
 $V_s = 200mV$
 $10k\Omega$
 $1\mu A$

$I_C = \frac{10V - 5V}{1k\Omega} = \frac{5V}{1k\Omega} = 5mA$
 $I_C = \beta I_B$
 $5mA = 125 \cdot I_B$
 $I_B = 40\mu A$
 $V_T = 0.026V$
 $r_n = \frac{V_T}{I_B} = \frac{0.026V}{40\mu A} = 650\Omega$
 $g_m = \frac{I_C}{V_T} = \frac{5mA}{0.026V} = 192.5 \frac{mA}{V}$

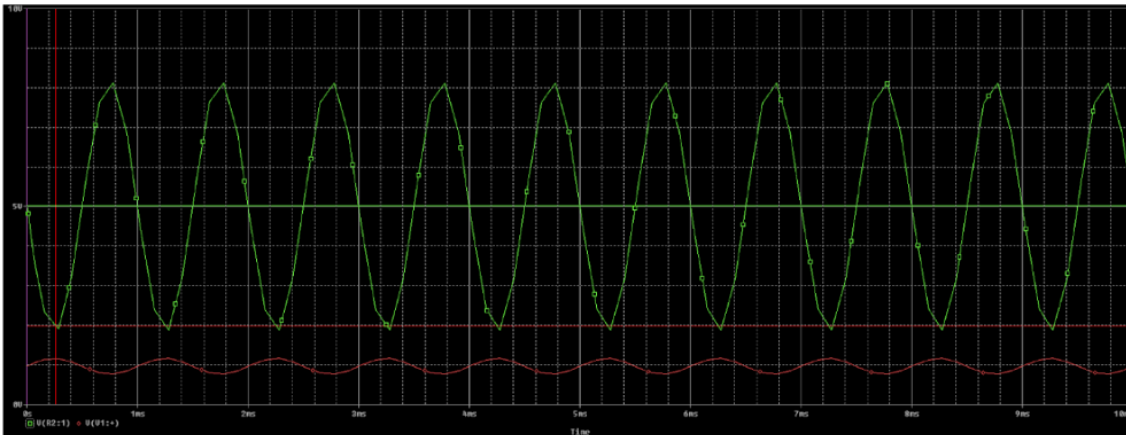
Small Signal

$V_{gs} = V_{sig} \left(\frac{650\Omega}{10k\Omega + 650\Omega} \right)$
 $V_{gs} = V_{sig} \cdot 0.061$
 $V_o = -g_m V_{gs} 1k\Omega$
 $V_o = -192.5 \frac{mA}{V} \cdot V_{sig} \cdot 0.061 \cdot 1k\Omega$
 $V_o = -192.5 \cdot 0.061 V_{sig}$
 $V_o = -11.73 V_{sig}$
 $V_o = -11.73 \cdot 200mV$
 $V_o = -2.35V$

$V_{cc} = 10V$
 $V_{cc} = 10V$



	Trace Color	Trace Name	Y1	Y2	Y1 - Y2	Y1(Cursor1) - Y2(Cursor2)	-3.0389			
		X Values	261.014u	0.000	261.014u	Y1 - Y1(Cursor1)	Y2 - Y2(Cursor2)	Max Y	Min Y	Avg Y
	CURSOR 1,2	V(R2:1)	1.9877	5.0265	-3.0389	0.000	0.000	5.0265	1.9877	3.5071
		V(V1:~)	1.1569	968.571m	188.276m	-830.826m	-4.0580	1.1569	968.571m	1.0627



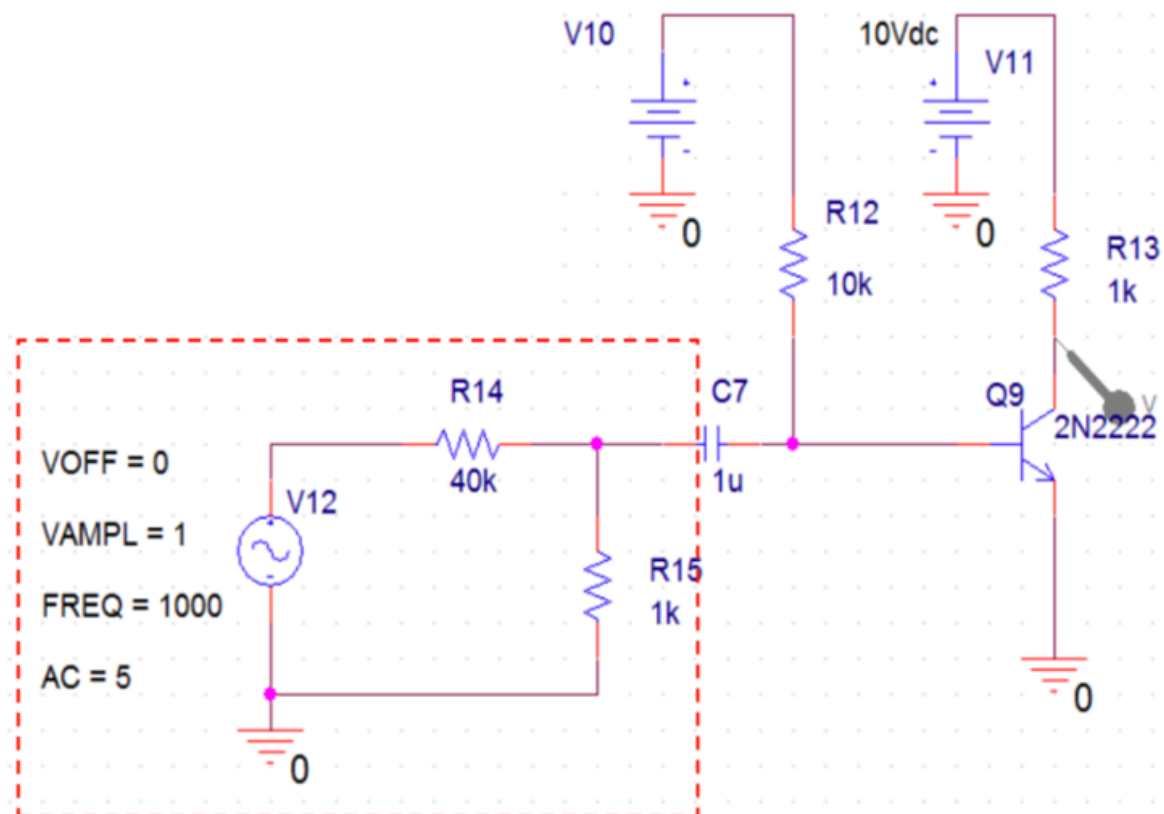
Conclusion: I believe that this experiment was a success due to the close correlation between measured, simulated, and calculated.

Experiment 3:

Experiment#3- Investigate a Common Emitter Amplifier (Connect a small signal source directly to the base terminal)

In this experiment investigate dc offset using an alternative Approach.

1. Construct the circuit and verify that the collector current is around $I_C = 5 \text{ mA}$ and V_{CE} around 5 V. Use the V10 from Experiment 1.
2. Using the oscilloscope monitor the waveforms at the base and collector.
3. Record the input and output waveforms. Record the amplitudes of both the input and output waveforms.



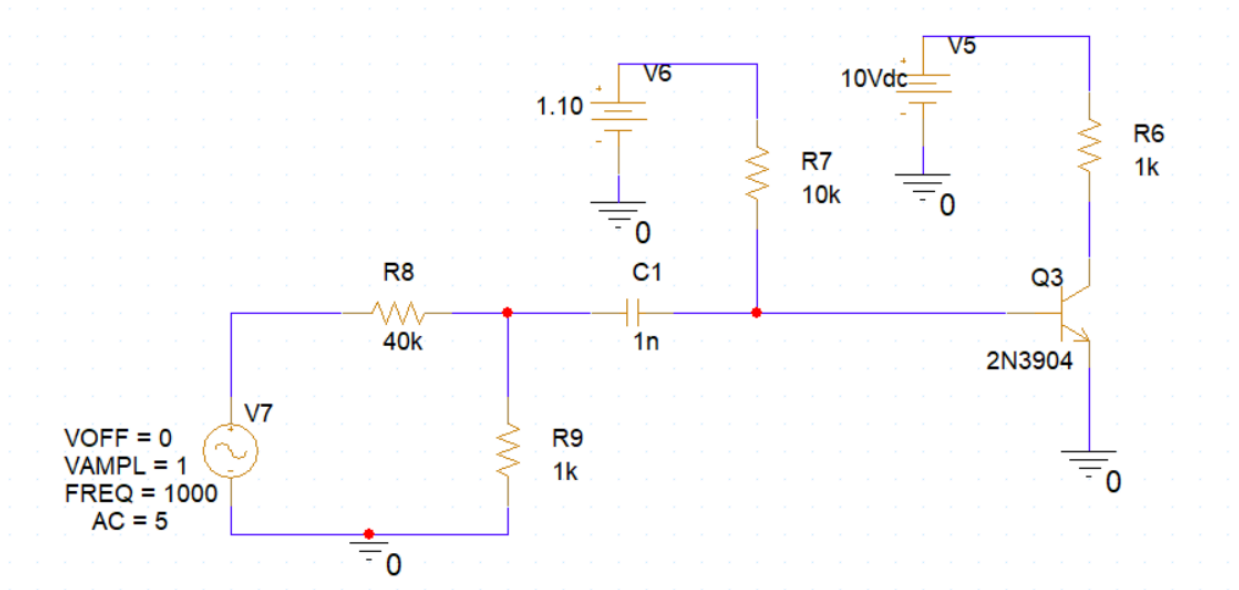
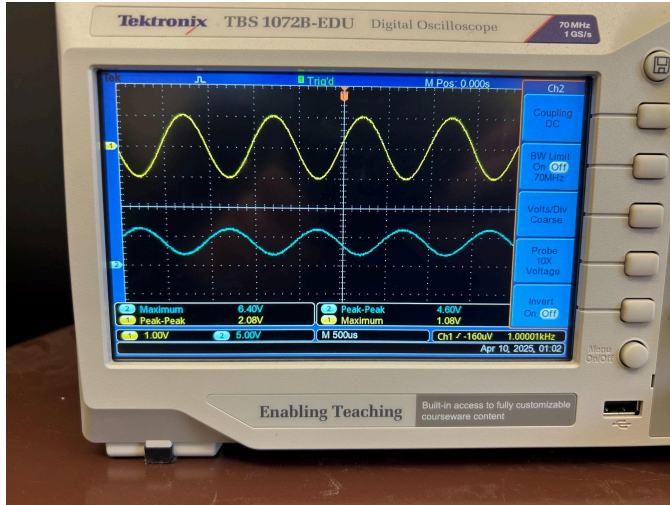
R12=9.9201KOHM

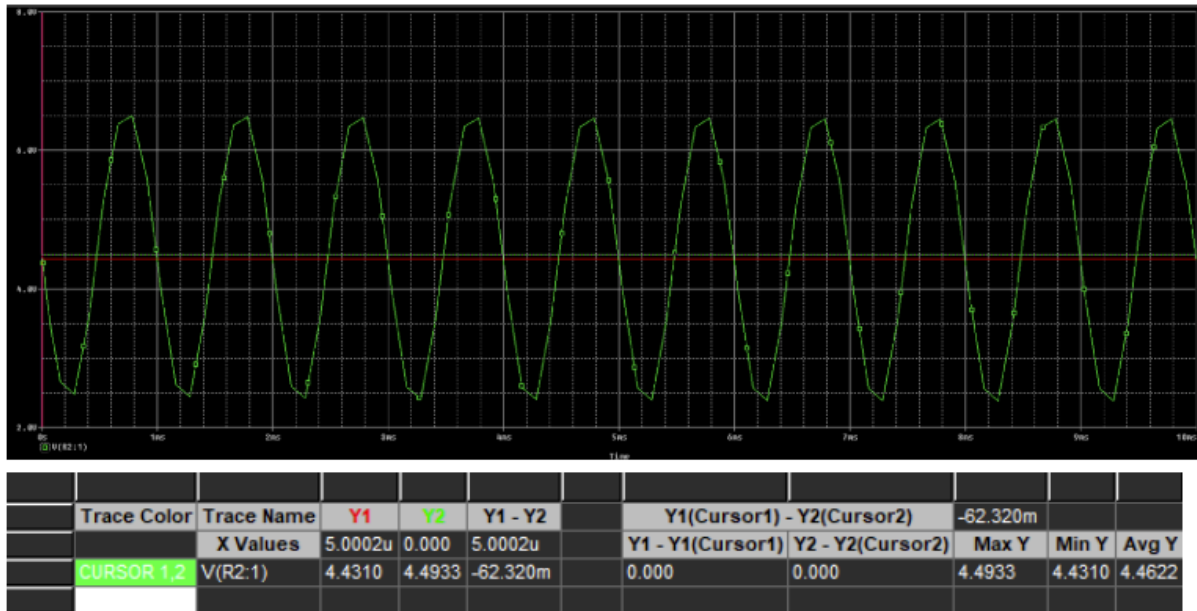
R13=0.9931KOHM

R14=38.851KOHM

R15=0.9898KOHM

C=1uF

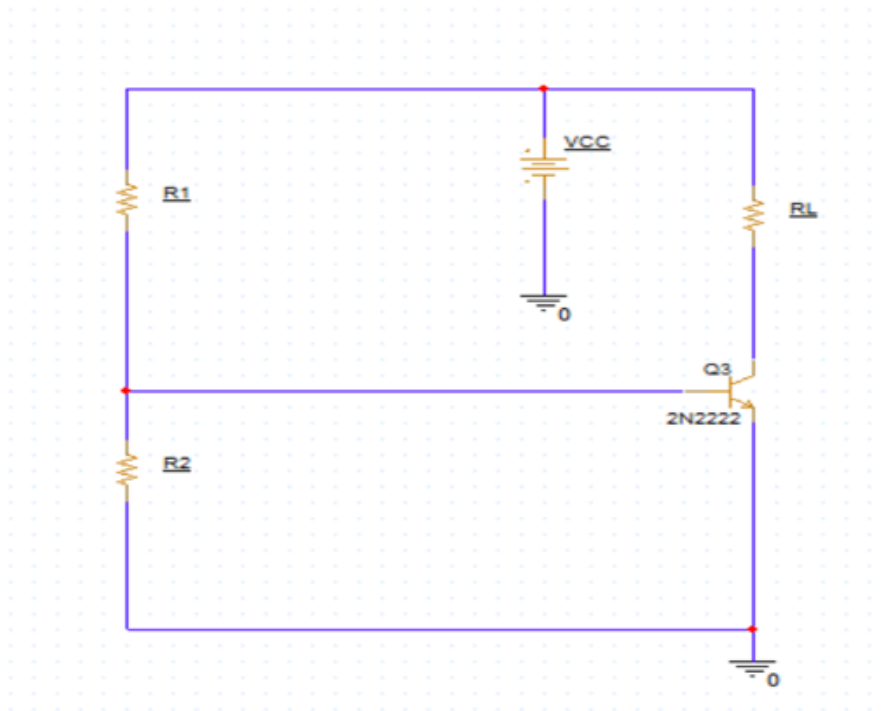




Conclusion: Again I believe that this experiment was a success due to the close correlation between simulated and measured.

Experiment 4:

Experiment #4: Investigate a Common Emitter Amplifier (with voltage divider biasing source)



1. Adjust the voltage divider resistors so output at the collector is around 5 V. Here $R_L = 1k$, $V_{CC} = 10V$. Record those settings. Record the amplitudes at the base (input) and collector (output).

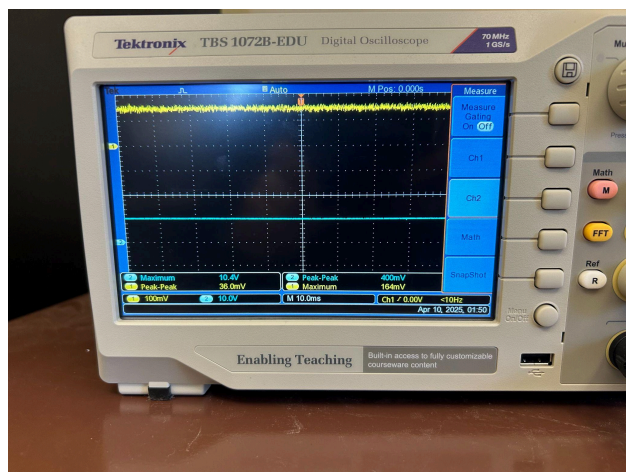
$R_1 = ?$

$R_2 = ?$

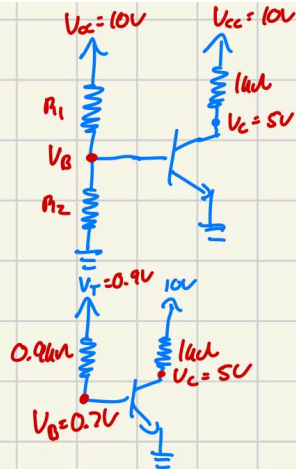
$R_L = 0.9931k\Omega$

$R_1 = 9.87563k\Omega$

$R_2 = 0.98867k\Omega$



Experiment 4:



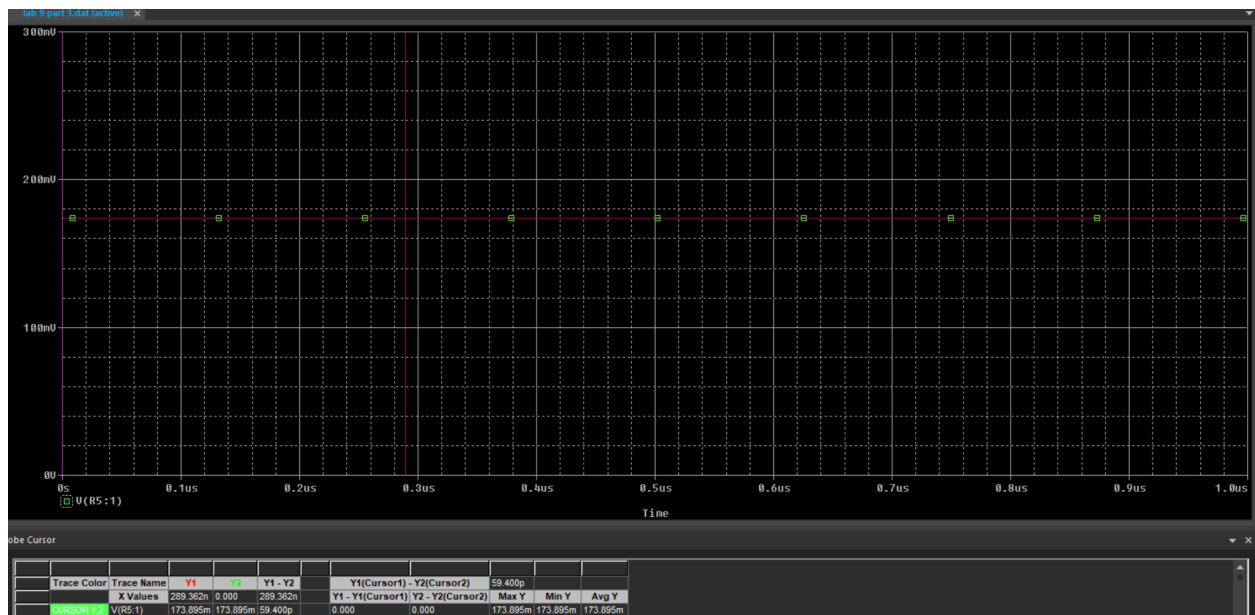
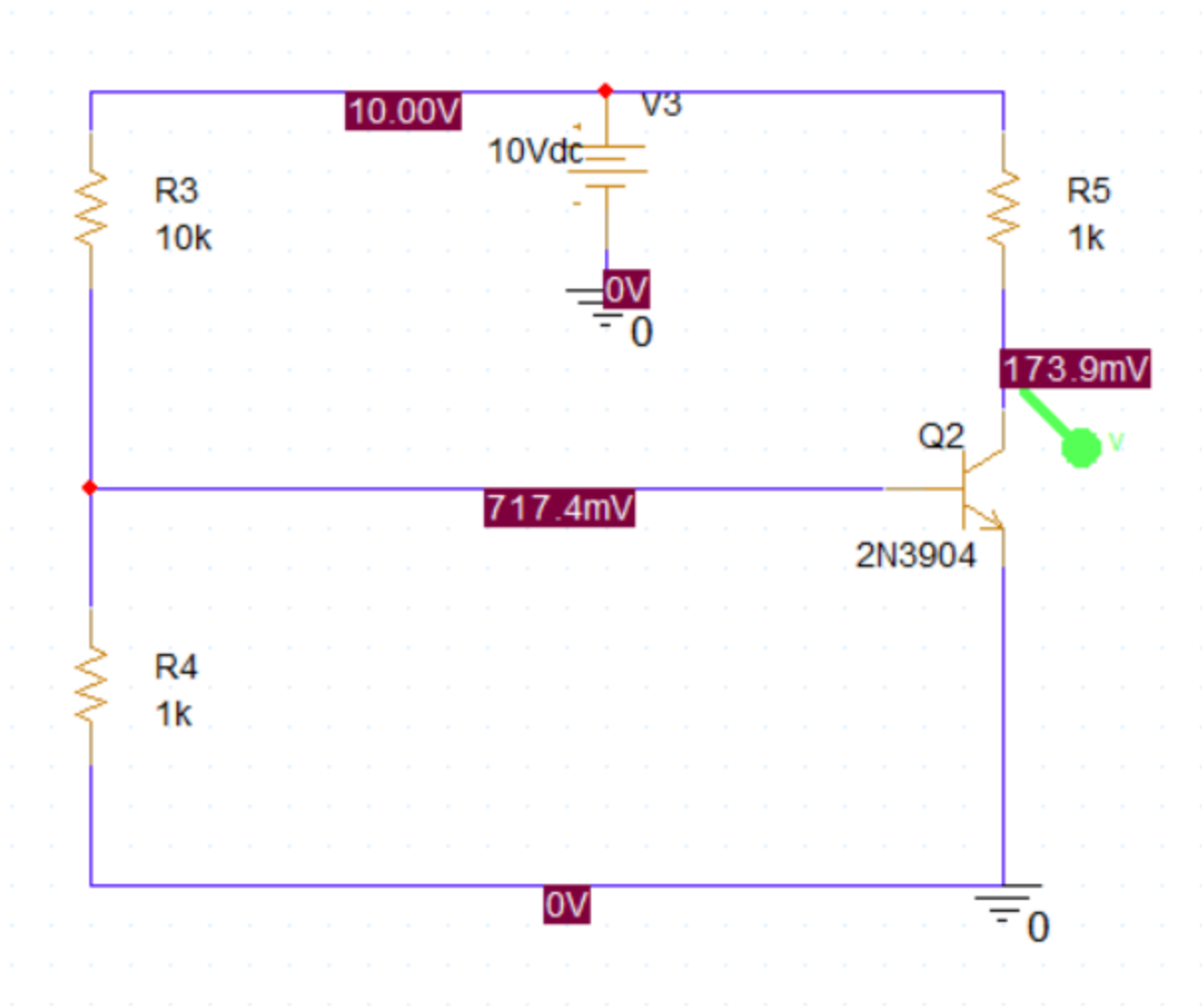
$$V_0 = -2.55V$$

$$V_T = V_{CC} \left(\frac{R_2}{R_1 + R_2} \right) = V_T = 10 \left(\frac{1k\Omega}{10k\Omega + 1k\Omega} \right) = V_T = 0.9V$$

$$R_{TT} = R_1 // R_2 = 10k\Omega // 1k\Omega = 0.9k\Omega$$

$$R_1 = 10k\Omega$$

$$R_2 = 1k\Omega$$



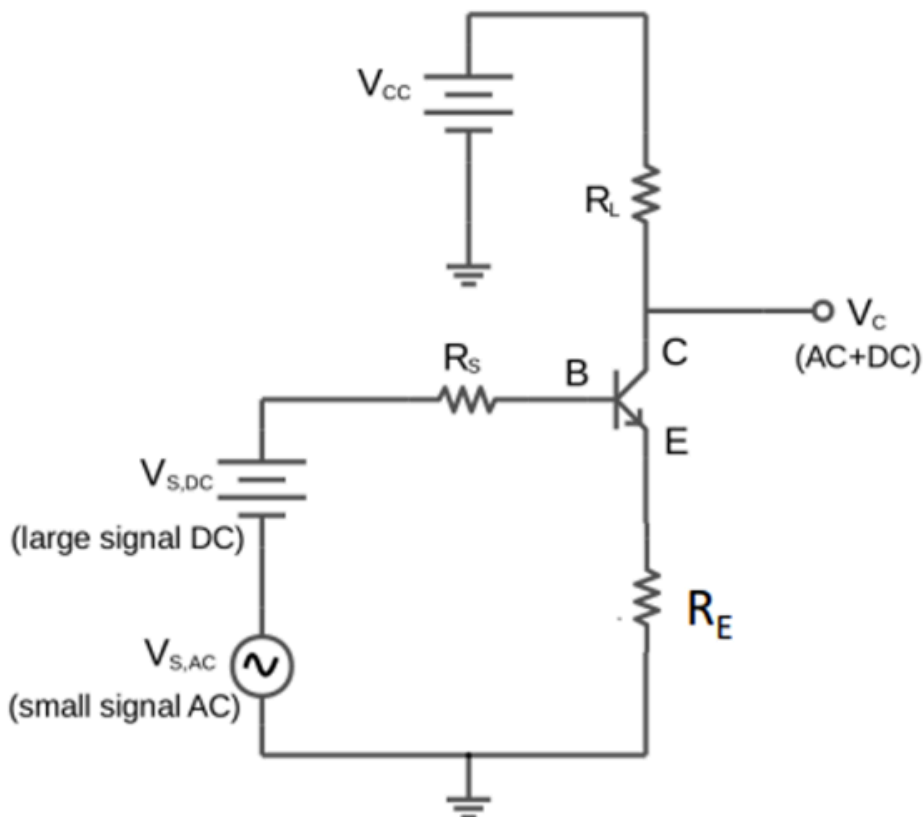
Conclusion: This experiment was a success due to the close correlation between the measured/simulated output waveforms.

Experiment 5:

Experiment #5: Investigate the following Common Emitter Amplifier

From the above experiment, now you have an idea in characterizing the BJT.

1. Use the circuit from experiment 2 and investigate the effect of adding a resistor R_E at the emitter in the range: $R_E = 100 \text{ ohm} - 1000 \text{ ohm}$
2. Record output waveforms and determine gain.



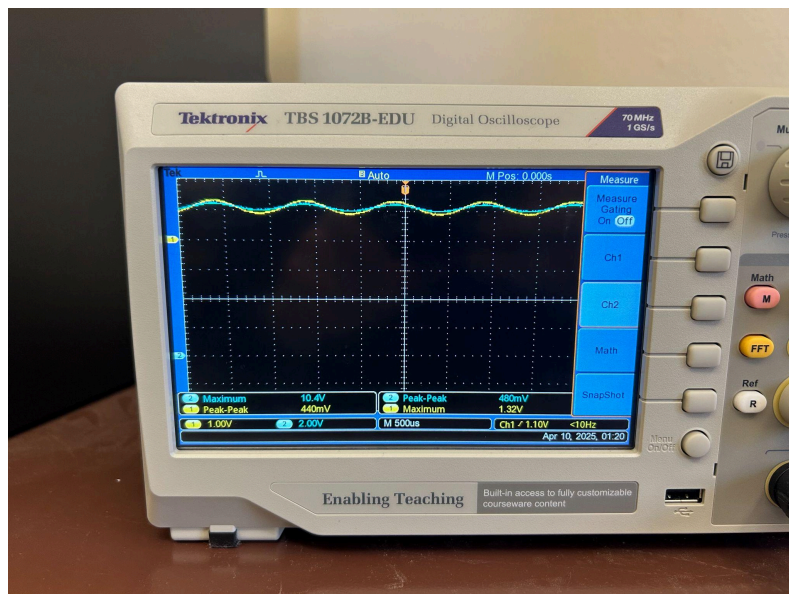
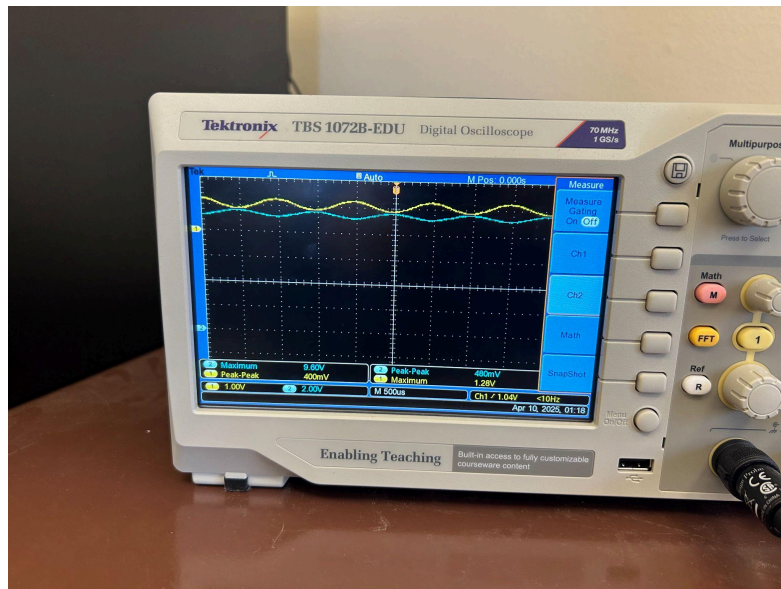
$R_L = 0.9931 \text{ KOHM}$

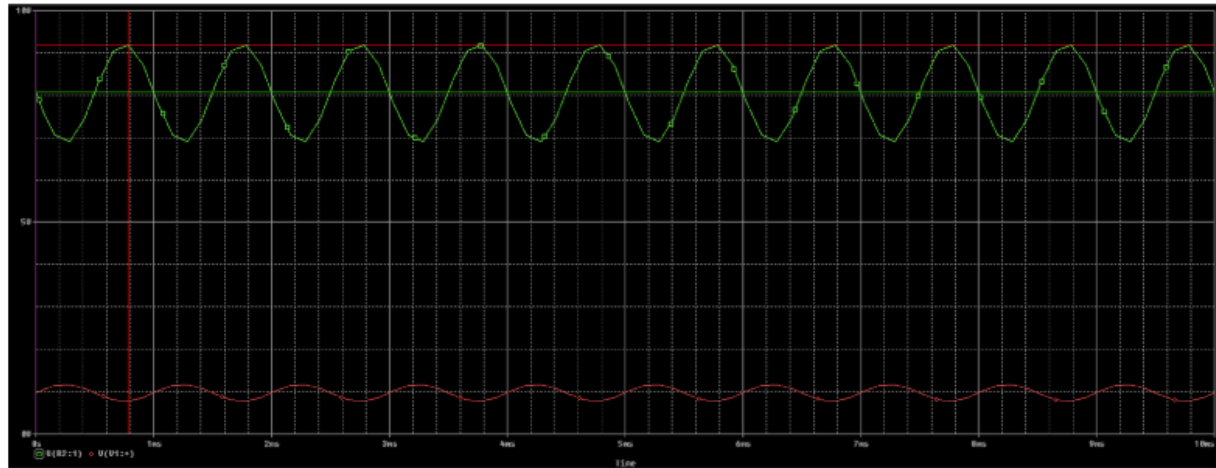
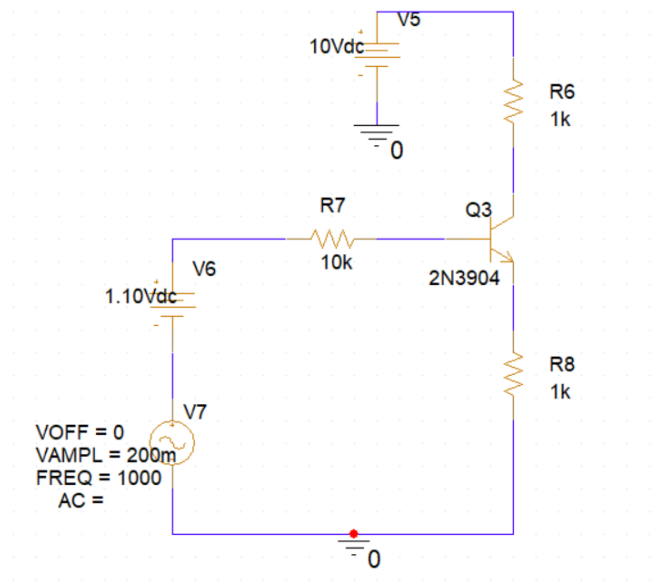
$R_S = 9.9201 \text{ KOHM}$

$V_{CC} = 10 \text{ V}$

$R_E = 0.387 \text{ KOHM}$ (1st Picture)

$R_E = 0.11950 \text{ KOHM}$ (2nd Picture)





Trace Color	Trace Name	Y1	Y2	Y1 - Y2	Y1(Cursor1) - Y2(Cursor2)	1.1047			
X Values		784.530u	0.000	784.530u	Y1 - Y1(Cursor1)	Y2 - Y2(Cursor2)	Max Y	Min Y	Avg Y
CURSOR 1,2	V(R2:1)	9.1797	8.0750	1.1047	0.000	0.000	9.1797	8.0750	8.6273
	V(V1:~)	771.920m	968.571m	-196.651m	-8.4078	-7.1064	968.571m	771.920m	870.246m

Conclusion: I believe that this experiment was a success due to the close relationship with experiment 2. Also all measured, simulated, and calculated values have a close correlation.